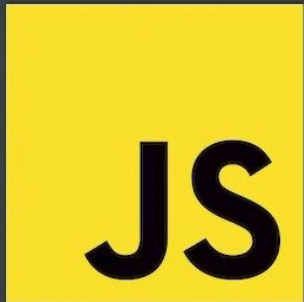


Javascript

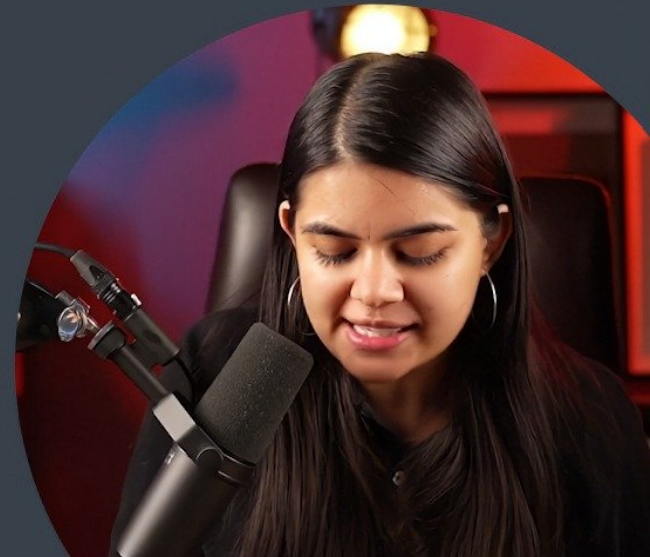
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Using the Console →

Uses **REPL**

Read-Evaluate-Print-Loop



Using the Console → vs code

Uses REPL

Read-Evaluate-Print-Loop



.html
.css
.js } ✓

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Using the Console → vs code

Uses REPL

Read-Evaluate-Print-Loop

↑
↑
↑
computes
calculations

clear → cmd + K
ctrl + d

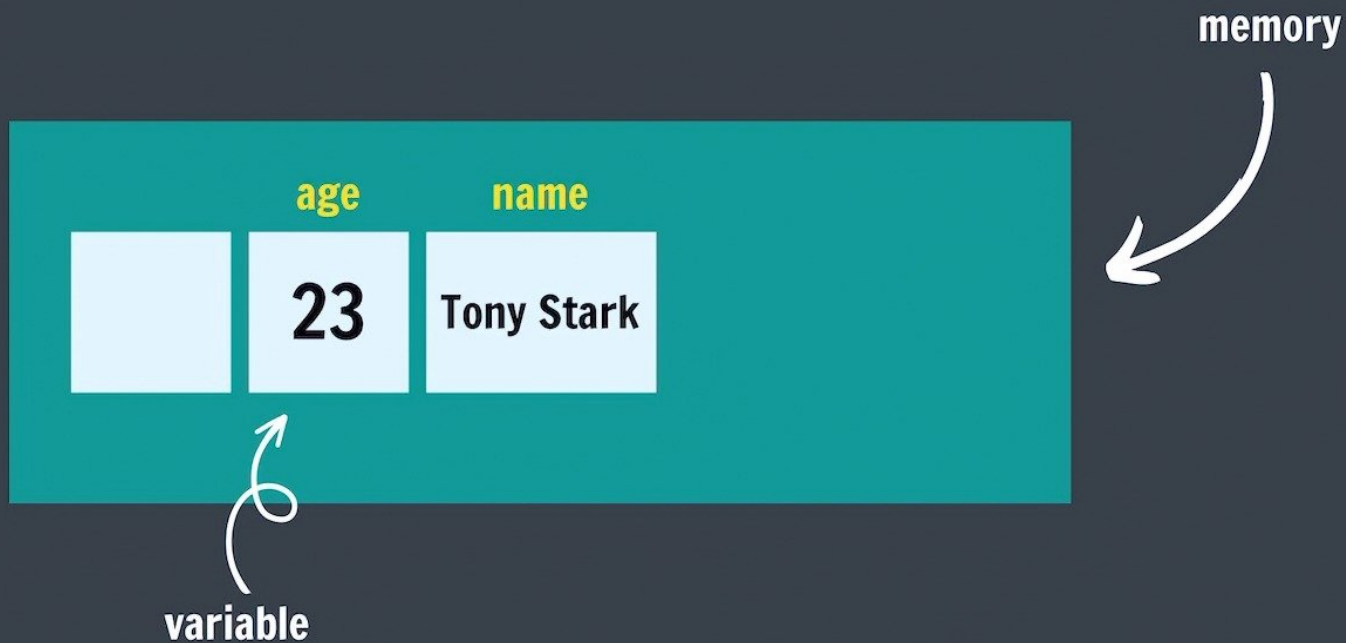
.html
.css
.js } ✓

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What is a Variable?

A variable is simply the name of a storage location.



What is a **Variable**?

A variable is simply the name of a storage location.



$a = 10$
~~age~~ = 23
↓ ↓
var val
name = "Tony"



New Tab

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Elements

Console

Sources

Network

>>

3

Settings

More

Close

top

Filter

Default levels

Settings

3 Issues: 3

< 10

> b = 5

< 5

> age = 23

< 23

> name = "tony"

< 'tony'

> a

< 10

> a + b

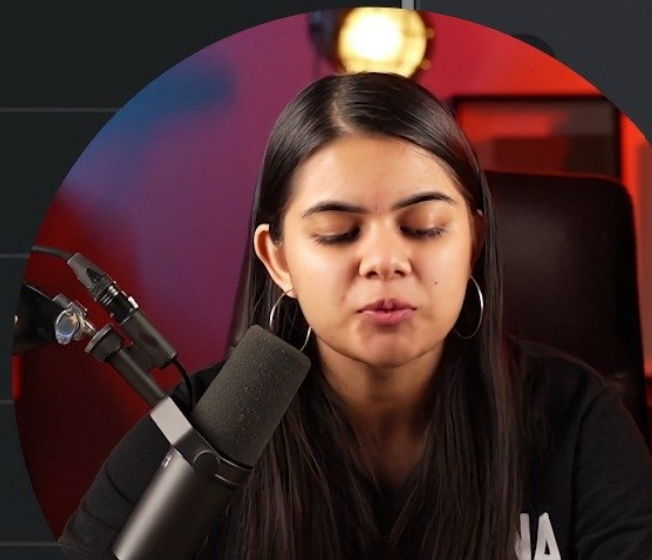
< 15

Go

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07:22 / 07:38



Data Types in JS

Primitive Types

- Number
- Boolean
- String
- Undefined
- Null
- BigInt
- Symbol

Handwritten notes:
age = 23
names = "tony" ← text
true / false

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follow -> yes or no? (**boolean**)



text (**string**)

count (**number**)



Numbers in JS

- Positive (14) & Negative (-4)
- Integers (45, -50)
- Floating numbers - with decimal (4.6, -8.9)



Numbers in JS

- Positive (14) & Negative (-4)
- Integers (45, -50)
- Floating numbers - with decimal (4.6, -8.9)

C++ } type
C }
Java }
float → decimal
int → +ve -ve

a = 14 ✓
a = -4 ✓

a = 4.6 ✓
a = -4.6 ✓
a = 0 ✓



Operations in JS

```
a = 20
```

```
b = 10
```

```
//addition
```

```
sum = a + b
```

```
//subtraction
```

```
diff = a - b
```

```
//multiplication
```

```
prod = a * b
```

```
//division
```

```
div = a / b
```

```
//modulo
```

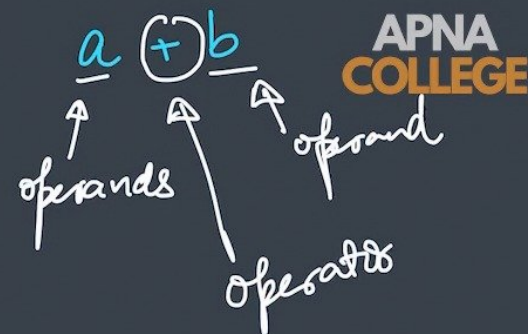
```
rem = a % b
```

- Modulo (remainder operator)

$$12 \% 5 = 2$$

- Exponentiation (power operator)

$$2^{**}3 = 8$$



Operations in JS

```
a = 20
```

```
b = 10
```

```
//addition
```

```
sum = a + b
```

```
//subtraction
```

```
diff = a - b
```

```
//multiplication
```

```
prod = a * b
```

```
//division
```

```
div = a / b
```

```
//modulo
```

```
rem = a % b
```

- Modulo (remainder operator)

$$12 \% 5 = 2$$

- Exponentiation (power operator)

$$2^{**}3 = 8$$

$$\underline{a^{**}b} =$$

$$\underline{\text{odd}} \quad \underline{\text{odd}} \% 2 = 1$$

$$\underline{\text{even}} \quad \underline{\text{even}} \% 2 = 0$$

$$2^2 = 4 = 2 * 2$$

$$2^4 = \underbrace{2 * 2 * 2 * 2}_{4 \times} = 16$$

a^b



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Seat Matrix 2022

Department of Co...

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Elements

Console

Sources

Network

Performance

>>

3

top

Filter

Default levels

3 Issues: 3

> 2**3

< 8

> a = 3

< 3

> b = 3

< 3

> a ** b

< 27

> 0 / 0

< NaN

> typeof NaN

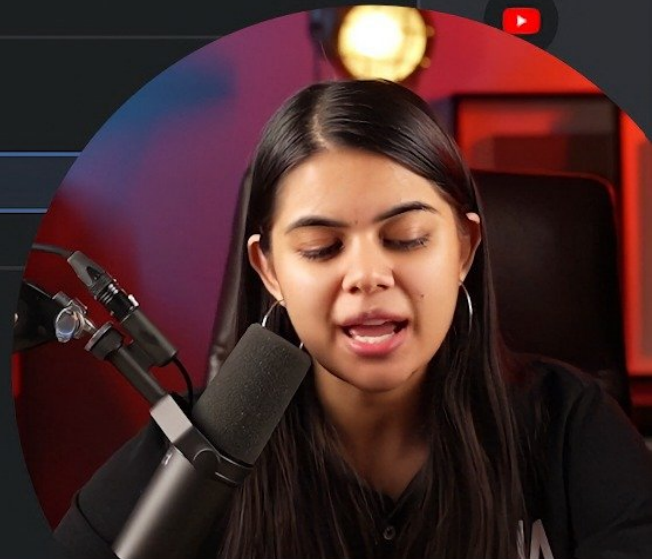
< 'number'

>

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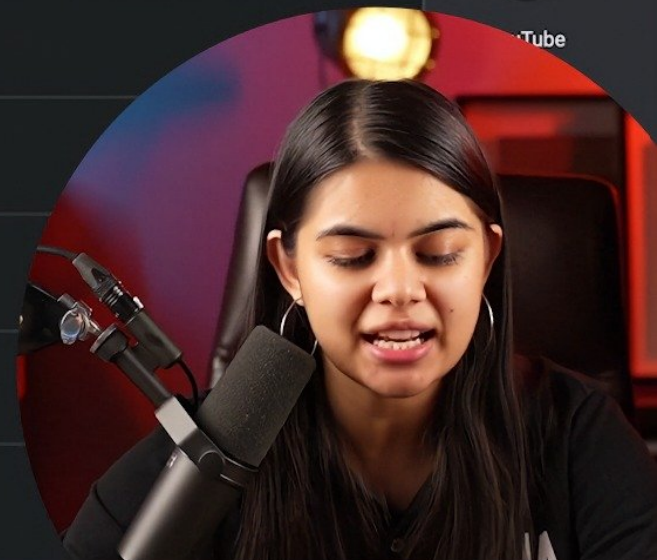
Go

Search Google



```
> 2**3  
< 8  
> a = 3  
< 3  
> b = 3  
< 3  
> a ** b  
< 27  
> 0 / 0  
< NaN  
> typeof NaN  
< 'number'  
> NaN + 1  
< NaN  
> NaN - 1  
< NaN  
> |
```

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NaN in JS

The NaN global property is a value representing **Not-A-Number**.

0 / 0

NaN - 1

NaN * 1

NaN + NaN

1 1
2 2
2.5 2.5
NaN
=

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Operator Precedence

This is the general **order** of solving an expression.

()

**

*, /, %

+, -

$$5 + 4 * 10 / 2 + 8$$

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BODMAS



Operator Precedence

This is the general **order** of solving an expression.

()
↑
**
↑
*, /, %
↑
+, -

$$5 + 4 * 10 / 2 + 8$$

APNA5 COLLEGE

BODMAS

$$(5 + 2) / 7 + 1 * 2$$
$$(7) / 7 + 1 * 2$$
$$1 + 2$$



Operator Precedence

Evaluate

$(2 + 1) * 3$

$3 / 1 + 2 ** 2$

$4 + 1 * 6 / 2$



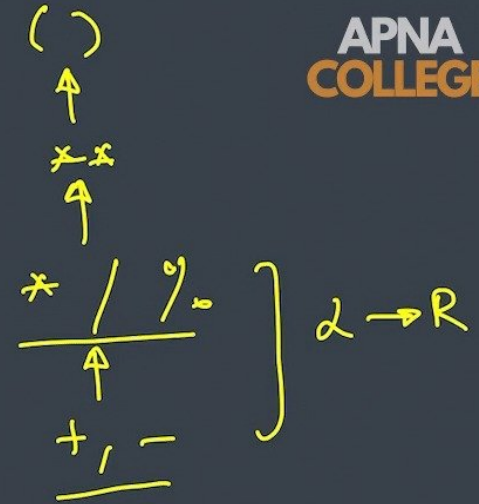
Operator Precedence

Evaluate

$$(2 + 1) * 3 \longrightarrow 3 * 3 \longrightarrow 9$$

$$3 / 1 + 2 ** 2 \longrightarrow 3 / 1 + 4 \longrightarrow 3 + 4 \longrightarrow 7$$

$$4 + 1 * 6 / 2 \longrightarrow 4 + 6 / 2 -$$



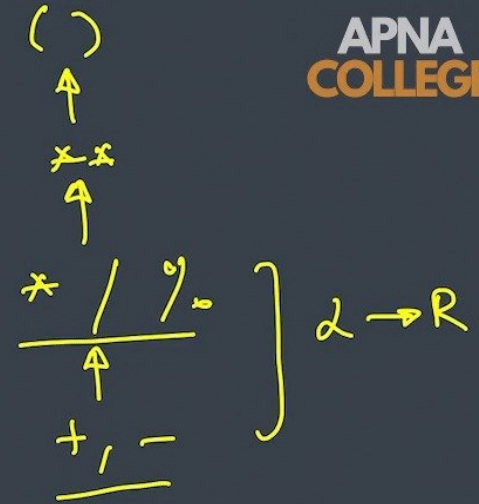
Operator Precedence

Evaluate

$$(2 + 1) * 3 \longrightarrow 3 * 3 \longrightarrow 9$$

$$3 / 1 + 2 * 2 \longrightarrow 3 / 1 + 4 \longrightarrow 3 + 4 \longrightarrow 7$$

$$4 + 1 * 6 / 2 \longrightarrow 4 + 6 / 2 \longrightarrow 4 + 3 \longrightarrow \underline{\underline{7}}$$



$$\underline{a} = \underline{5}$$

let keyword

Syntax of declaring variables

```
let age = 23 ;
```

```
age = age + 1 ;
```

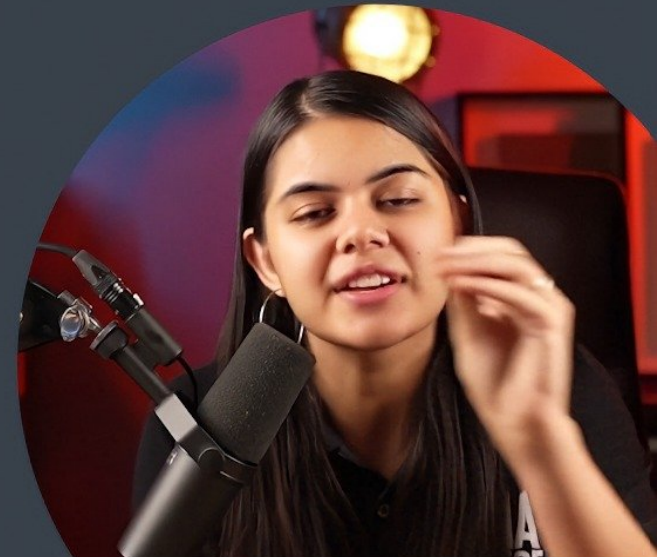
```
let cgpa ;
```

```
cgpa = 8.9
```

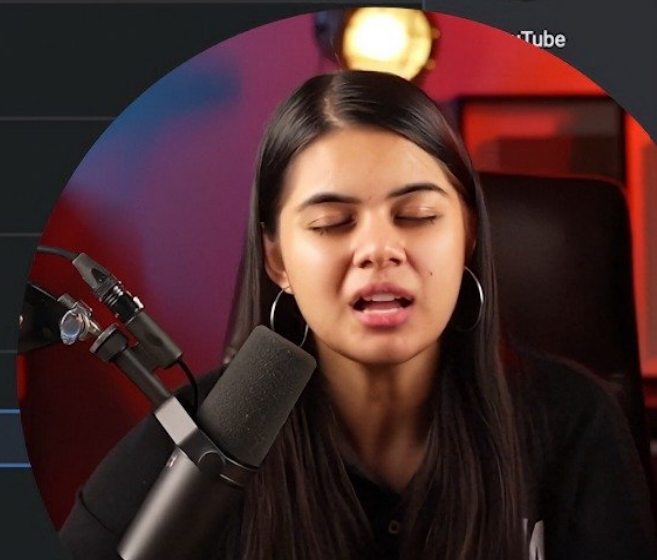
```
let num1 = 1 ;
```

```
let num2 = 2 ;
```

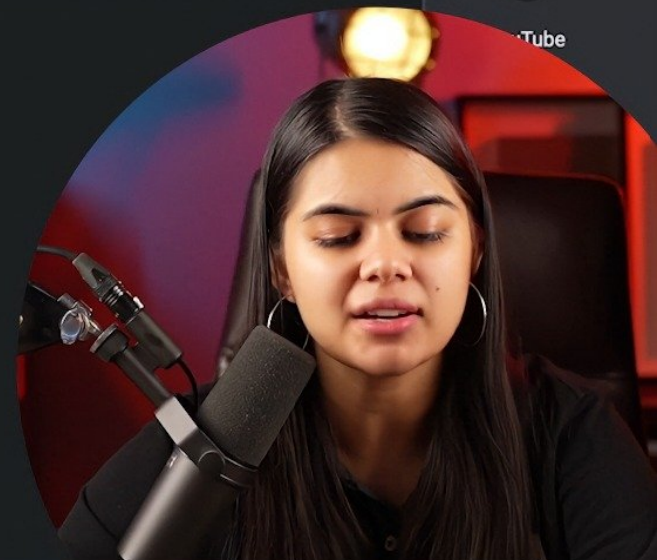
```
let sum = num1 + num2 ;
```



```
> age = age + 1;
< 24
> age
< 24
> age = age - 1;
< 23
> age
< 23
> let cgpa = 8.9;
< undefined
> cgpa
< 8.9
> cgpa = cgpa * 10;
< 89
> cgpa
< 89
>
```



```
> let num1 = 1;
< undefined
> let num2 = 2;
< undefined
> let finalSum = num1 + num2;
< undefined
> finalSum
< 3
>
```



const keyword

values of constants can't be changed with re-assignment & they can't be re-declared

```
const year = 2025;
```

```
year = 2026      // Error
```

```
year = year + 1  // Error
```

```
const pi = 3.14 ;
```

```
const g = 9.8 ;
```



```
> const newNum = 5;
```

```
< undefined
```

```
> newNum
```

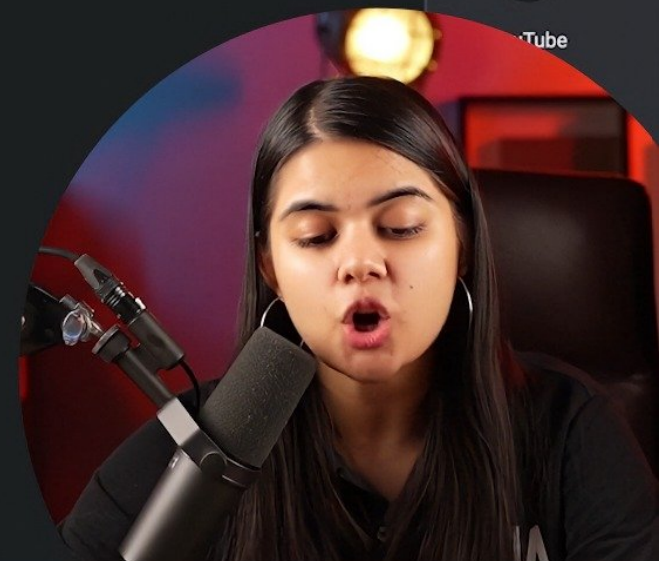
```
< 5
```

```
> newNum = 10;
```

```
✖ ▶ Uncaught TypeError: Assignment to constant variable.  
    at <anonymous>:1:8
```

VM3697:1

```
>
```



```
> const pi = 3.14;
< undefined

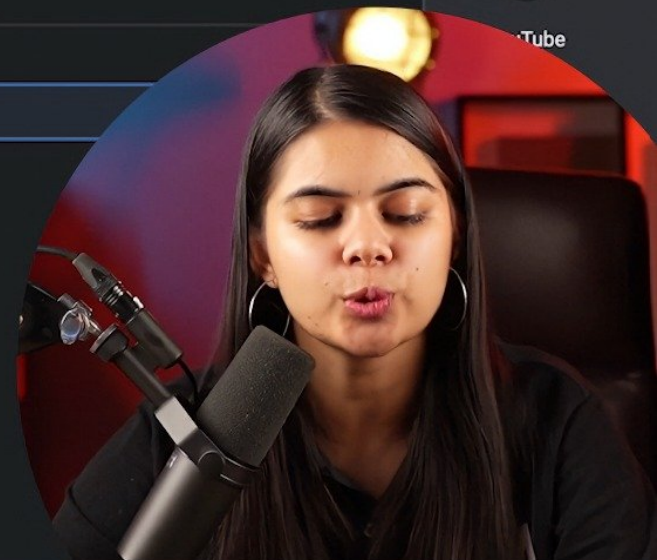
> let rad = 4;
< undefined

> let area = pi * r ** 2;
✖ ▶ Uncaught ReferenceError: r is not defined
  at <anonymous>:1:17 VM3887:1

> let area = pi * rad ** 2;
< undefined

> area
< 50.24

>
```



var keyword

Old Syntax of writing variables

```
var age = 23 ; ✓
```

```
var cgpa = 8.9 ; ✓
```

```
var num1 = 1 ;
```

```
var num2 = 2 ;
```

```
var sum = num1 + num2 ;
```



Practice Qs

Qs. What is the value of age after this code runs?



```
let age = 23 ;  
age + 2; //after 2 years
```

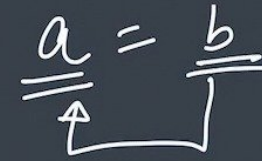
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Qs. What is the value of avg after this code runs?

```
let hindi = 80;  
let eng = 90;  
let math = 100;  
let avg = (hindi + eng + math) / 3;
```



Assignment Operators

$$\underline{\underline{a}} = \underline{\underline{b}}$$


age = age + 1

age += 1

age = age - 1

age -= 1

age = age * 1

age *= 1



Unary Operators

`age = age + 1`

`age += 1`

`age++` // increment

`age = age - 1`

`age -= 1`

`age--` // decrement

binary operators
2 operands

$a + b$

$a \times b$

a / b



$++a$ (pre-inc)
 $a++$ (post-inc)

Unary Operators

Pre-increment (Change, then use)

```
let age = 10 ;
```

```
let newAge = ++age ;
```

Post-increment (Use, then change)

```
let age = 10 ;
```

```
let newAge = age++ ;
```



Unary Operators

$++a$ (pre-inc)
 $a++$ (post-inc)

$--a$ (pre-dec)
 $a--$ (post-dec)

Pre-increment (Change, then use)

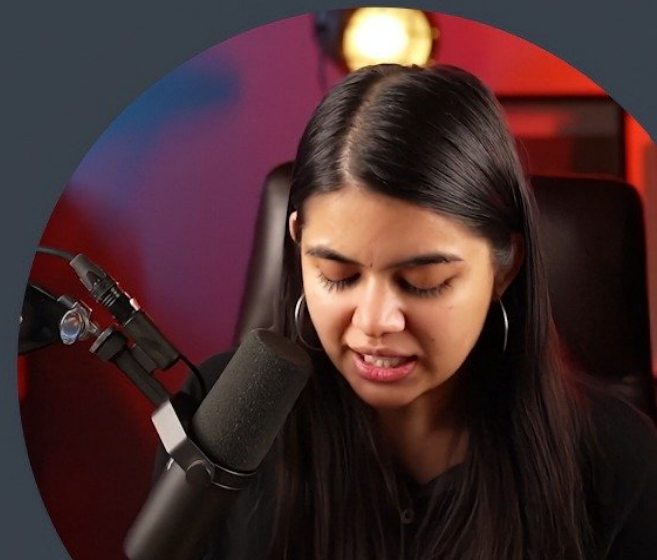
```
let age = 10 ;
```

```
let newAge = ++age ;
```

Post-increment (Use, then change)

```
let age = 10 ;
```

```
let newAge = age++ ;
```



Unary Operators

$++a$ (pre-inc)
 $a++$ (post-inc)

$--a$ (pre-dec)
 $a--$ (post-dec)

Pre-increment (Change, then use)

```
let age = 10 ;
```

```
let newAge = ++age ;
```



~~age = 10~~ ||
newAge = 11

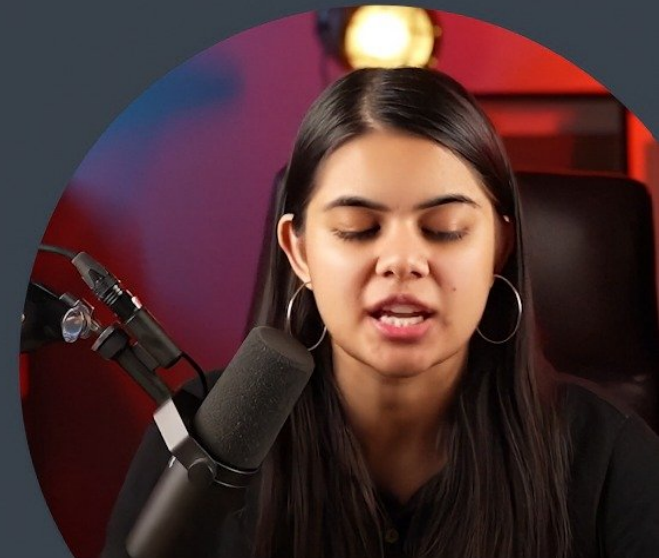
Post-increment (Use, then change)

```
let age = 10 ;
```

```
let newAge = age++ ;
```



~~age = 10~~ ||
newAge = 10



Practice Qs

Qs. What is the value of each variable in each line of code?

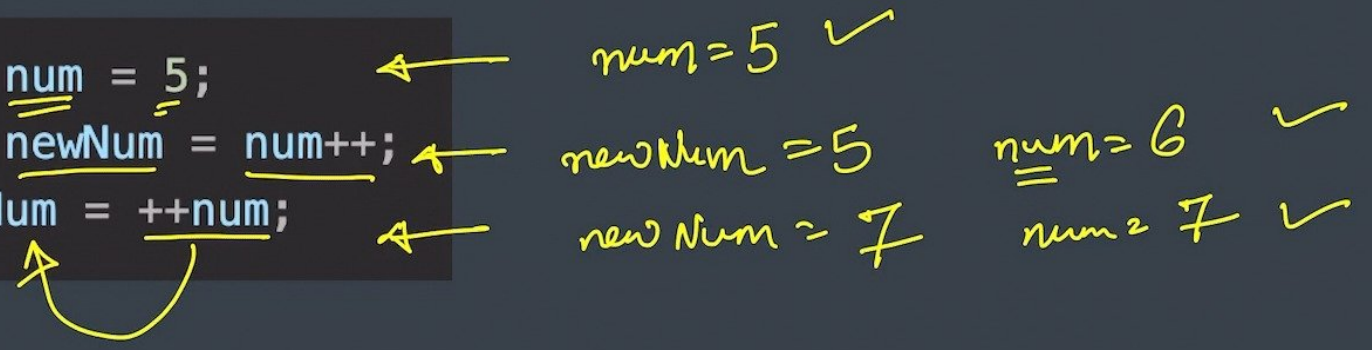
```
let num = 5;
let newNum = num++;
newNum = ++num;
```



Practice Qs

Qs. What is the value of each variable in each line of code?

```
let num = 5;  
let newNum = num++;  
newNum = ++num;
```



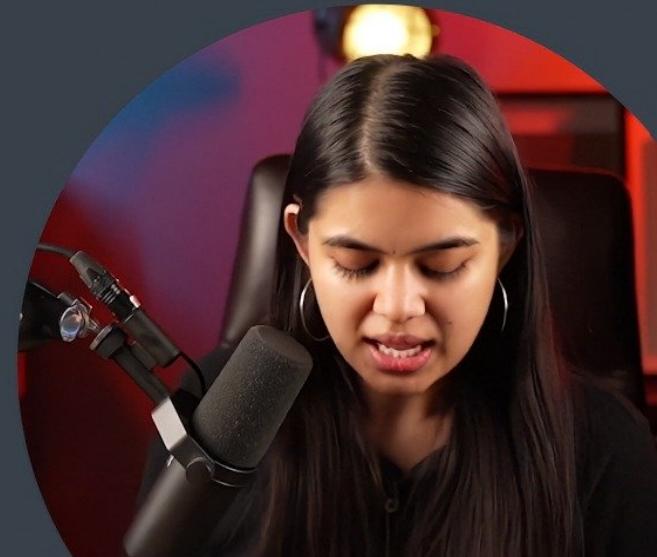
num = 5 ✓

newNum = 5

num = 6 ✓

newNum = 7

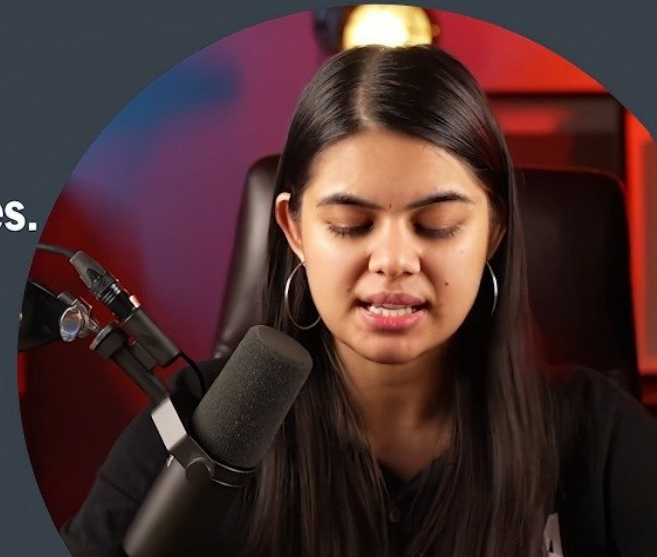
num = 7 ✓



Identifier Rules ✓

All JavaScript variables must be identified with **unique names** (identifiers).

- Names can contain letters, digits, underscores, and \$ dollar signs. (no space)
A-Z a-z 0-9 _ \$
- Names must begin with a letter.
- Names can also begin with \$ and _.
- Names are case sensitive (y and Y are different variables).
- Reserved words (like JavaScript keywords) CANNOT be used as names.



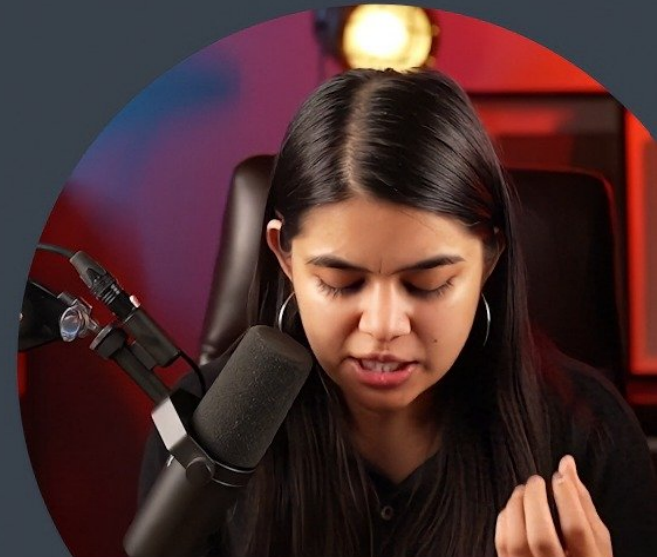
camelCase

Way of writing identifiers

cameCase (**JS naming convention**)

snake_case

PascalCase



camelCase

Way of writing identifiers

camelCase (JS naming convention)
↔↔↔

camelCase

let totalPrice

let fullName;

snake_case

let full_name;

PascalCase

let FullName;

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```
> let age = 13;
< undefined

> let isAdult = false;
< undefined

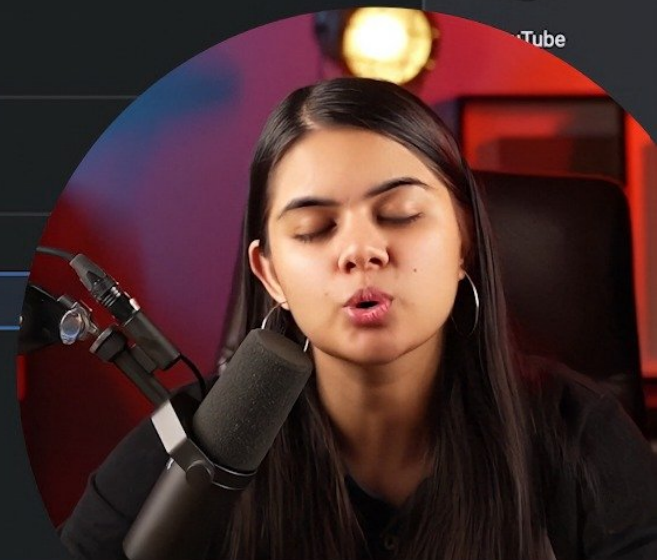
> age = 25;
< 25

> isAdult = true;
< true

> isAdult
< true

> isAdult = false;
< false

> isAdult
< false
>
```



Boolean in JS

[Java
C++]

[let a = 5
a = true]

ADITYA
COLLEGE
boolean

Boolean represents a truth value -> **true or false** / yes or no

```
let age = 23 ;
```

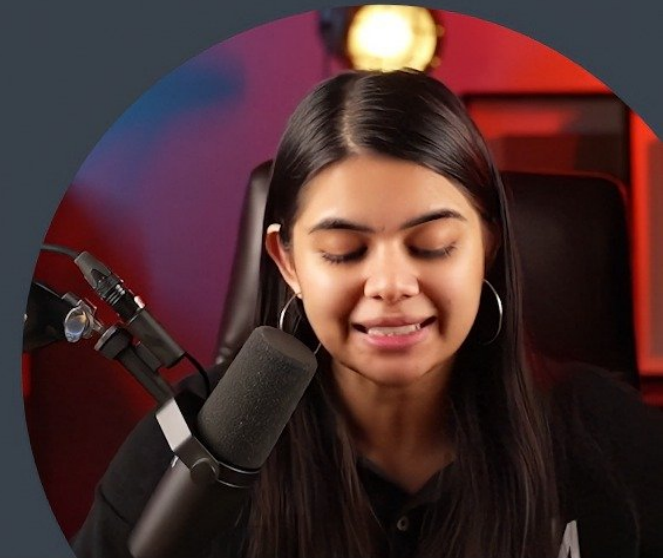
```
let isAdult = true ;
```

var type change

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```
let age = 13 ;
```

```
let isAdult = false ;
```



What is TypeScript?

Static Typed, where JS is dynamic typed

Designed by Microsoft



TS
a = 5 ✓
a = true X
a = 10 ✓

JS



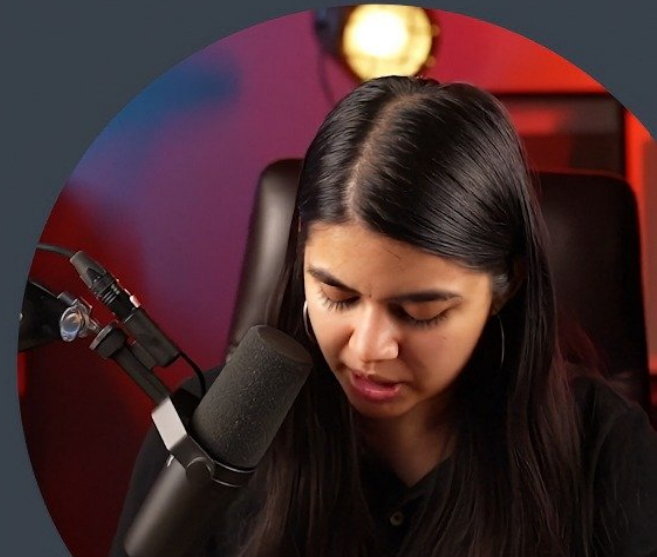
Practice Qs

Qs. Find the errors in the following code ?

a) `let ①age = 5;` ← *let age1 = 5;*
`let ②age = 10;` ← *let age2 = 10;*

b) `let marks = 75;` ←
`let isPass = True;` ←
true

c) `let isPass = ①true';` }
boolean



String in JS

Strings are **text** or sequence of characters
→ word, page

```
let name = "Tony Stark";
```

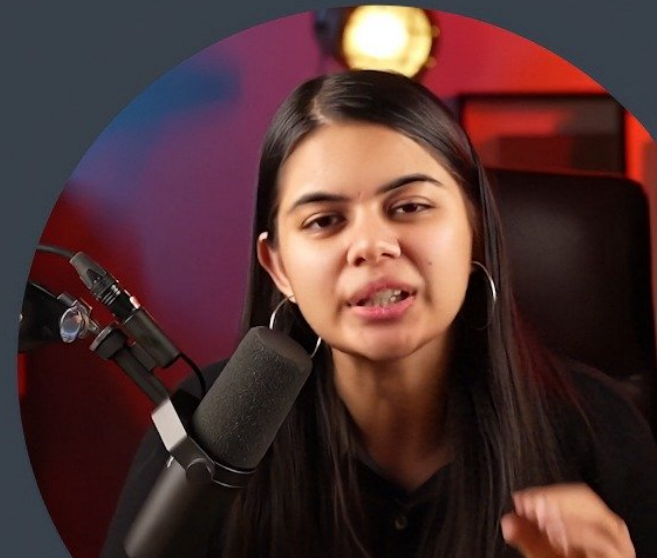
```
let role = 'ironman';
```

```
let char = 'a';
```

```
let num = '23';
```

```
let empty = " ";
```

“ { } ”
‘ ’ ,




```
> let firstName = "Tony";
```

```
< undefined
```

```
> firstName
```

```
< 'Tony'
```

```
> typeof firstName
```

```
< 'string'
```

```
> let lastName = 'Stark';
```

```
< undefined
```

```
> lastName
```

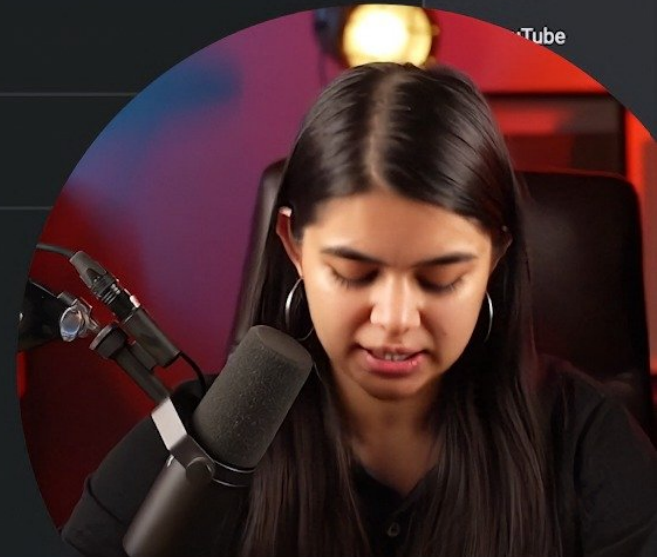
```
< 'Stark'
```

```
> typeof lastName
```

```
< 'string'
```

```
>
```

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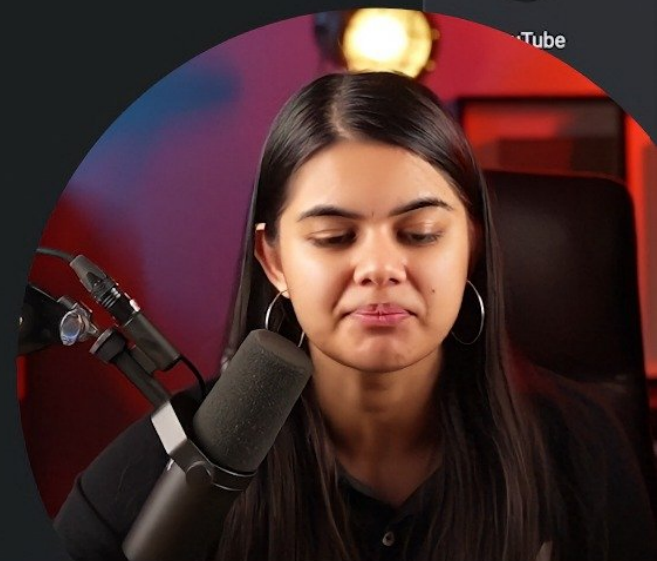


```
> let str = "svjadvjabdjvbjdfbvj      asdmfksdnfgsdfg"
< undefined

> str = "asfnaskjdfnjksd&##89349821374";
< 'asfnaskjdfnjksd&##89349821374'

> str = "asfnaskjdfnjksd&##89349821374asdfasf      23412zssdf";
< 'asfnaskjdfnjksd&##89349821374asdfasf      23412zssdf'

> str
< 'asfnaskjdfnjksd&##89349821374asdfasf      23412zssdf'
>
```



this is apple

String in JS

Strings are **text** or sequence of characters
→ word, phrase

```
let name = "Tony Stark";
```

```
let role = 'ironman';
```

```
let char = 'a';
```

```
let num = '23';
```

```
let empty = " ";
```

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```
> let sentence = 'this is "apple"';  
< undefined  
> sentence  
< 'this is "apple"'  
>
```

armankhan082020@gmail.com

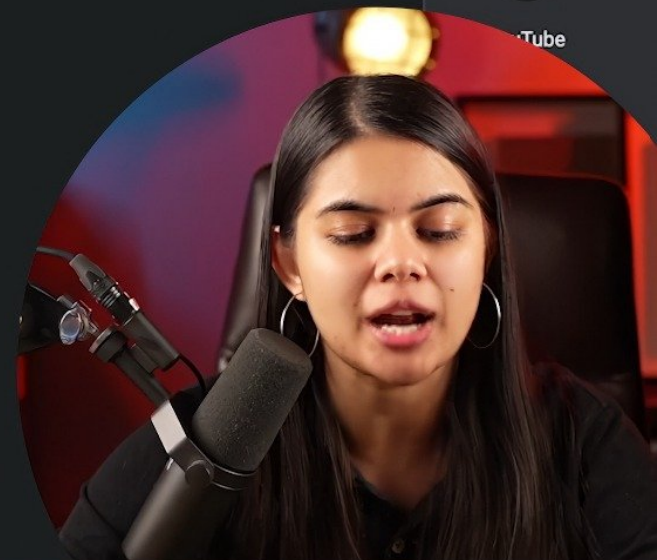
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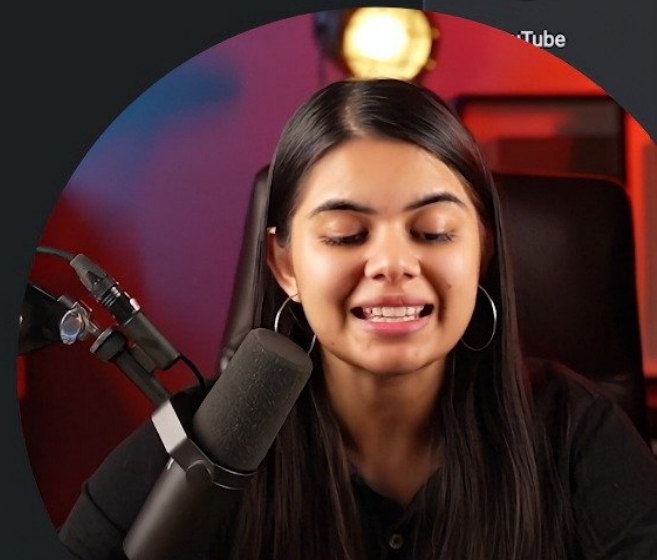


Tube




```
> let sentence = 'this is "apple"';  
< undefined  
> sentence  
< 'this is "apple"'  
> let sentence = "this is 'apple'";  
< undefined  
> sentence  
< "this is 'apple'"  
> |
```

armankhan082020@gmail.com



```
> let sentence = 'this is "apple"';  
< undefined  
> sentence  
< 'this is "apple"'  
> let sentence = "this is 'apple'";  
< undefined  
> sentence  
< "this is 'apple'"  
> let sentence = 'this is 'apple';
```

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Uncaught SyntaxError: Unexpected identifier 'apple'

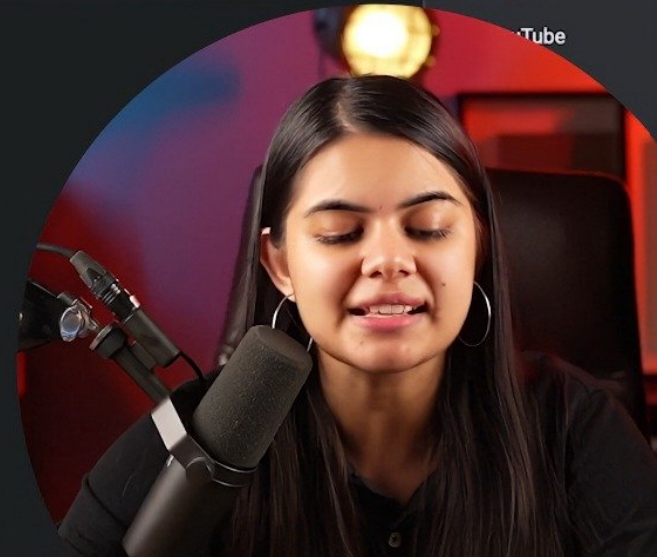
VM7234:1

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Go

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Tube



String Indices

index = position
0 based indexing

```
let name = "TONY STARK";
```

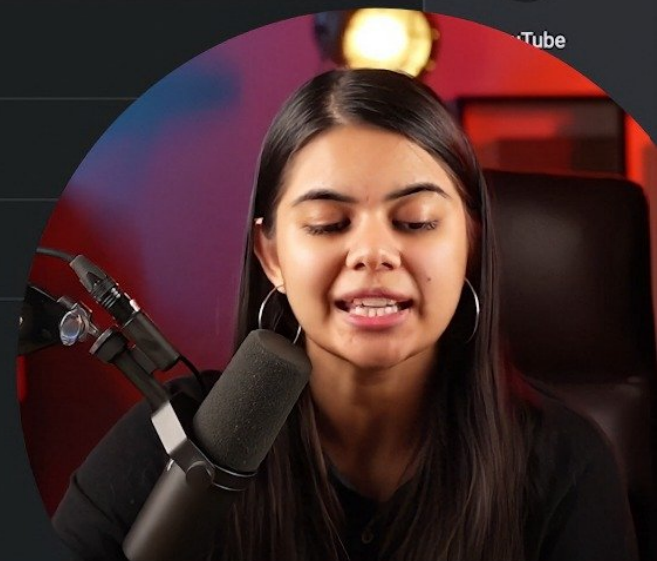
<u>T</u>	<u>O</u>	<u>N</u>	<u>Y</u>	<u> </u>	<u>S</u>	<u>T</u>	<u>A</u>	<u>R</u>	<u>K</u>
<u>index:</u> 0	1	2	3	4	5	6	7	8	9

name[0] -> 'T'

name[1] -> 'O' ...




```
> let name = "TONY STARK";
< undefined
> name
< 'TONY STARK'
> name[0]
< 'T'
> name[1]
< 'O'
> name[2]
< 'N'
> name[3]
< 'Y'
> name[4]
< ' '
> name[1000]
< undefined
>
```




```
> let name = "TONY STARK";
```

```
< undefined
```

```
> name
```

```
< 'TONY STARK'
```

```
> name[0]
```

```
< 'T'
```

```
> name[1]
```

```
< 'O'
```

```
> name[2]
```

```
< 'N'
```

```
> name[3]
```

```
< 'Y'
```

```
> name[4]
```

```
< ' '
```

```
> name[1000]
```

```
< undefined
```

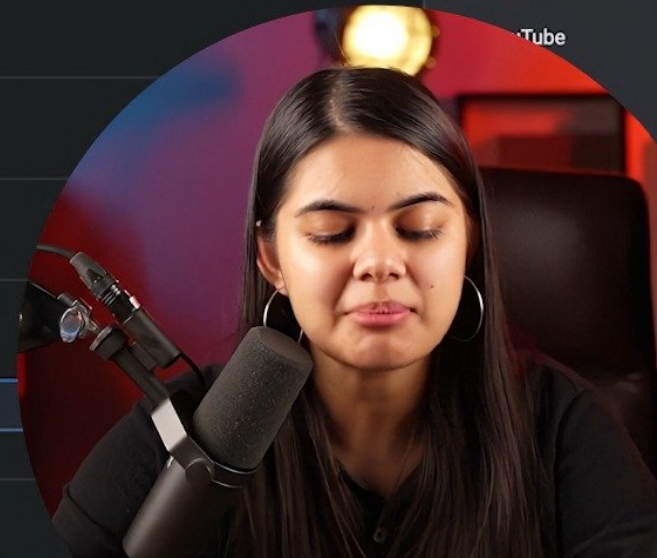
```
> name.length
```

```
< 10
```

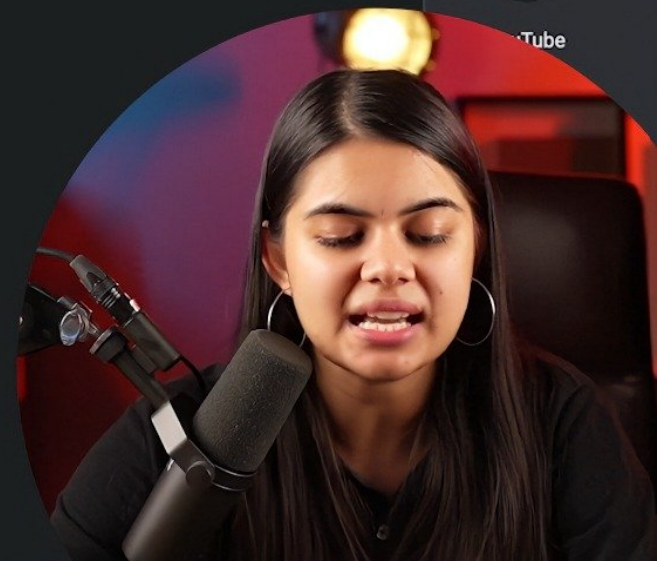
```
> typeof name.length
```

```
< 'number'
```

```
>
```



```
> let name = "TONY STARK";  
< undefined  
  
> name[name.length];  
< undefined  
  
> name[name.length-1];  
< 'K'  
  
> name[name.length-1];  
< 'K'
```



Concatenation

adding strings together

`"tony" + " " + "stark" = "tony stark"`

`"tony" + 1 = "tony1"`

armankhan082020@gmail.com



```
> "tony" + " " + "stark"
< 'tony stark'

> let firstName = "tony";
< undefined

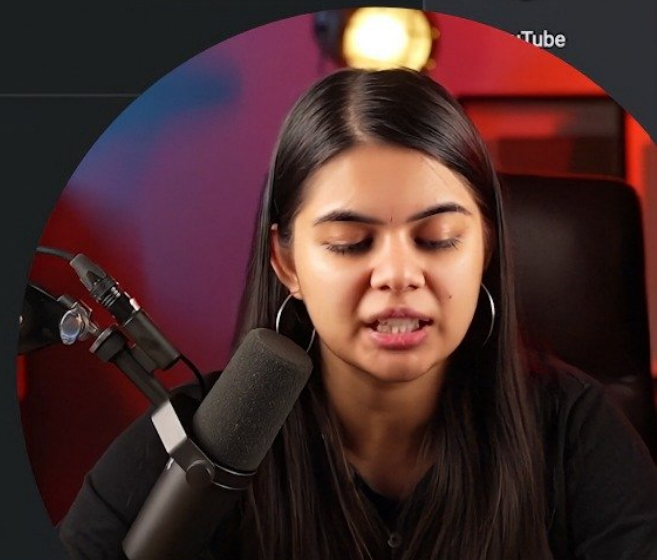
> let lastName = "stark";
< undefined

> let fullName = firstName + lastName;
< undefined

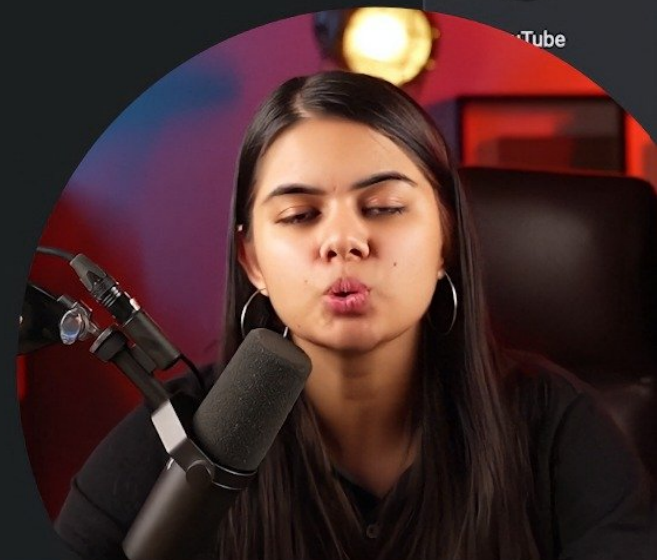
> fullName
< 'tonystark'

> fullName = firstName + " " + lastName;
< 'tony stark'

> fullName = firstName + " " + lastName;|
```




```
> let name;  
< undefined  
  
> name  
< undefined  
  
> typeof name  
< 'undefined'  
>
```



null & undefined in JS

undefined

A variable that has not been assigned a value is of type undefined.

```
> let a;  
↵ undefined  
  
> a  
↵ undefined
```

null

The null value represents the **intentional** absence of any object value.

To be explicitly assigned.

```
> let a = null;  
↵ undefined  
  
> a  
↵ null
```



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```
> let a = undefined;
< undefined

> a
< undefined

> let a;
< undefined

> let a = null;
< undefined
```

armankhan082020@gmail.com

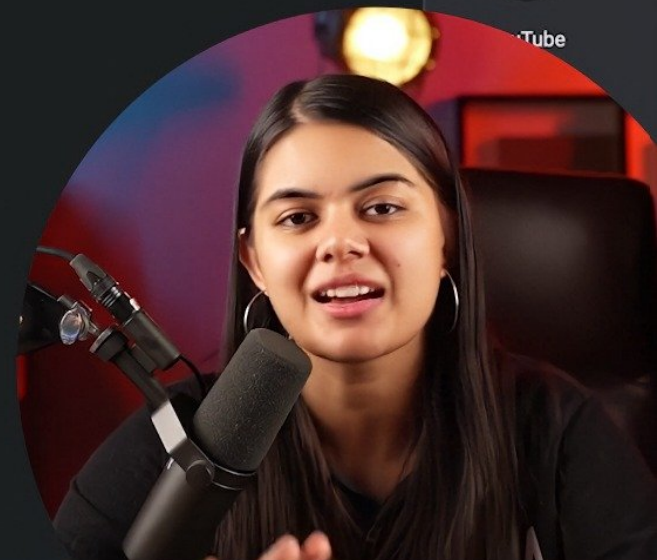
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```
> let a = undefined;
```

```
< undefined
```

```
> a
```

```
< undefined
```

```
> let a;
```

```
< undefined
```

```
> let a = null;
```

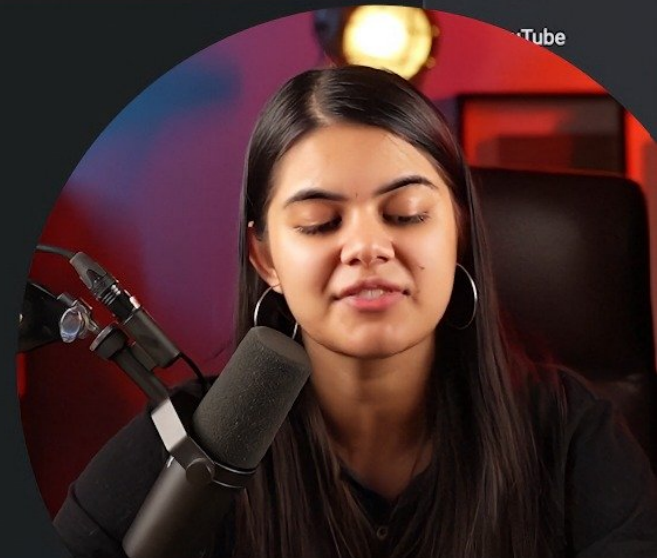
```
< undefined
```

```
> let null = 4;
```

```
✖ Uncaught SyntaxError: Unexpected token 'null'
```

VM8430:1

```
>
```



Practice Qs

Qs. Declare your name as a string and print its length in JS.

Qs. Declare your first name as a string and print its first character.

Qs. Declare your first name as a string and print its last character.

Qs. What is output of following code :

```
"apnacollege"+123|
```

Qs. What are lengths of an empty string & a string with a single space?



Practice Qs

Qs. Declare your name as a string and print its length in JS.

Qs. Declare your first name as a string and print its first character.

Qs. Declare your first name as a string and print its last character.

Qs. What is output of following code :

```
"apnacollege"+123
```

"apnacollege123"

Qs. What are lengths of an empty string & a string with a single space?

"" → 0
" " → 1
↑
space

